Sales & Profit Analysis Using Excel and Power BI

1. Introduction

This project aims to analyze business performance using modern data-analysis tools, starting with Excel and progressing to Power BI. The workflow includes data cleaning, preparation, exploratory analysis, and the development of three interactive dashboards that provide insights into sales, profitability, and geographic performance.

The final output supports data-driven decision-making by identifying top-performing products, regions, and customer segments.

2. Dataset Description

The dataset contains transactional sales records with detailed information, including:

- Category
- Sub-Category
- Region
- State
- City
- Sales
- Profit
- Profit Margin
- Discount
- Quantity
- Customer Type
- Segment
- Ship Mode
- Total Sales
- Total Profit
- Total Quantity

The cleaned dataset was stored in Excel and later imported into Power BI under the name:

cleaned_data

3. Excel Phase: Data Cleaning & Initial Analysis

3.1 Data Cleaning

The raw dataset was cleaned in Excel using the following steps:

- Removing duplicate records
- Standardizing category, sub-category, state, and city names
- Fixing inconsistent formatting
- Handling missing values
- Converting data types (Sales, Profit, Quantity)
- Creating calculated fields:
    - Total Sales
    - Total Profit
    - Total Quantity

3.2 PivotTable Analysis

An initial analytical dashboard was created using PivotTables to explore the data and validate its structure.

PivotTables included:

- Sales by Category
- Profit by Category
- Sales by Region
- Discount impact analysis
- Top Sub-Categories by performance

Purpose of the Excel Phase

To ensure data quality, extract early insights, and prepare the dataset for advanced visualization in Power BI.

4. Power BI Phase: Dashboard Development

After cleaning the data, the Excel table was imported into Power BI. Three interactive dashboards were created to analyze sales, profitability, and geographic performance.

4.1 Sales Performance Dashboard

This dashboard provides an overview of sales performance across categories, sub-categories, and regions.

Key Visuals

- Total Sales KPI
- Sales by Category
- Sales by Sub-Category
- Sales by Region
- Top 10 Products by Sales

Purpose

To identify which categories, sub-categories, and regions generate the highest sales.

4.2 Profitability Analysis Dashboard

This dashboard focuses on profit, discount impact, and margin performance.

Key Visuals

- Total Profit KPI
- Average Profit Margin
- Average Discount
- Profit by Category
- Discount vs Profit Scatter Plot
- Top 10 Most Profitable Sub-Categories
- Profit Margin by Category

Purpose

To understand profitability drivers and evaluate how discounts affect profit.

4.3 Geographic Sales & Profit Analysis Dashboard

This dashboard provides a geographic breakdown of sales and profit across states, regions, and cities.

Key Visuals

- Sales by State (Map)
- Profit by State (Shape Map)
- Sales by Region
- Profit by Region
- Top Cities by Sales
- Profit by Region and Category (Matrix)

Purpose

To identify high-performing locations, regional trends, and geographic opportunities.

5. Key Findings

Sales Insights

- The **Technology** category generates the highest sales.
- **Furniture** has the lowest sales performance.
- The **West** region leads in total sales.

Profit Insights

- **Technology** is the most profitable category.
- **Office Supplies** has the strongest profit margin.
- Higher discounts significantly reduce profitability.

Geographic Insights

- States such as **California** and **New York** show strong performance.
- The **West** region leads in both sales and profit.
- Cities like **Seattle**, **Los Angeles**, and **New York City** stand out as top performers.

6. Recommendations

1. Increase investment in the Technology category

It consistently delivers the highest sales and profit.

2. Review discount policies

High discounts negatively impact profitability and should be optimized.

3. Strengthen operations in the West region

It is the strongest region across all metrics.

4. Improve Furniture category performance

Consider pricing adjustments or supply-chain improvements.

5. Target high-performing cities

Focus marketing and inventory strategies on cities with strong demand.

7. Conclusion

This project demonstrates a complete analytical workflow, starting from Excel and ending with a professional multi-page Power BI dashboard. The insights gained help identify growth opportunities, optimize pricing strategies, and understand geographic trends that support strategic decision-making.

The combination of Excel and Power BI provides a powerful foundation for business analysis and visualization.

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